Find the no of vowels / consonants /spaces / special characters and digits in given string.

```
File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 1     Col 19  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h> #include<conio.h>
void main()
{
char  s[100]; int i,v,c,d,spa,spe;
clrscr();
printf("Enter the string ");gets(s);
for(v=d=c=spa=spe=i=0;s[i]!='\0';i++)
{
if(s[i]>='A' && s[i]<='Z') s[i]+=32; /* strlwr() */
if(s[i]>='a' && s[i]<='z')
{
if(s[i]=='a'||s[i]=='e'||s[i]=='i'||s[i]=='o'||s[i]=='u')v++;else c++;
}
else if(s[i]>='0' && s[i]<='9')d++;
else if(s[i]==' ')spa++;
else spe++;
}
printf("Vowels=%d, Consonants=%d, Digits=%d, spaces=%d, special=%d",
v,c,d,spa,spe);
getch();
}
```

```
Enter the string James Bond - 001
Vowels=3, Consonants=6, Digits=3, spaces=3, special=1_
```

```
for( i=0; s[i]!='\0';i++)
{
if( s[i]>='A' && s[i]<='Z') s[i]+=32; /* upper to lower */

if(s[i]>='a' && s[i]<='z')
{
if(s[i]=='a'||s[i]=='e'||s[i]=='i'||s[i]=='o'||s[i]=='u')v++;
else c++;
}
else if(s[i]>='0'&&s[i]<='9')d++;
else if(s[i]==' ')spa++;
else spe++;
}
p(v,c,d,spa,spe);
```

s	~~A~~ a	b	_	9	*	\0
	0	1	2	3	4	5

i	v	c	d	spa	spe
0	0	0	0	0	0
1	1	1			
2			1	1	1
3					
4					
5					

Using predefined functions:

```
#include<stdio.h> #include<conio.h>
#include<ctype.h>
void main()
{
char  s[100]; int i,v,c,d,spa,spe; clrscr();
printf("Enter the string ");gets(s);
for(v=d=c=spa=spe=i=0;s[i]!='\0';i++)
{
if(isalpha(s[i]))
{
s[i]=tolower(s[i]);
if(s[i]=='a'||s[i]=='e'||s[i]=='i'||s[i]=='o'||s[i]=='u')v++;else c++;
}
else if(isdigit(s[i]))d++;
else if(isspace(s[i]))spa++;
else spe++;
}
printf("Vowels=%d, Consonants=%d, Digits=%d, spaces=%d, special=%d",
v,c,d,spa,spe);
getch();
}
```

```
Enter the string Bahu Bali - 2
Vowels=4, Consonants=4, Digits=1, spaces=3, special=1
```

Replace lower/upper/digits/spaces/special with @/#/$/%/*

Using predefined functions:

TC

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Line 2 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char  s[100]; int i; clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++)
{
if(islower(s[i]))s[i]='@';
else if(isupper(s[i]))s[i]='#';
else if(isdigit(s[i]))s[i]='$';
else if(isspace(s[i]))s[i]='%';
else s[i]='*';
}
printf("Converted string %s",s);
getch();
}
```

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TC

```
Enter the string James Bond - 007
Converted string #@@@@%#@@@%*%$$$
```

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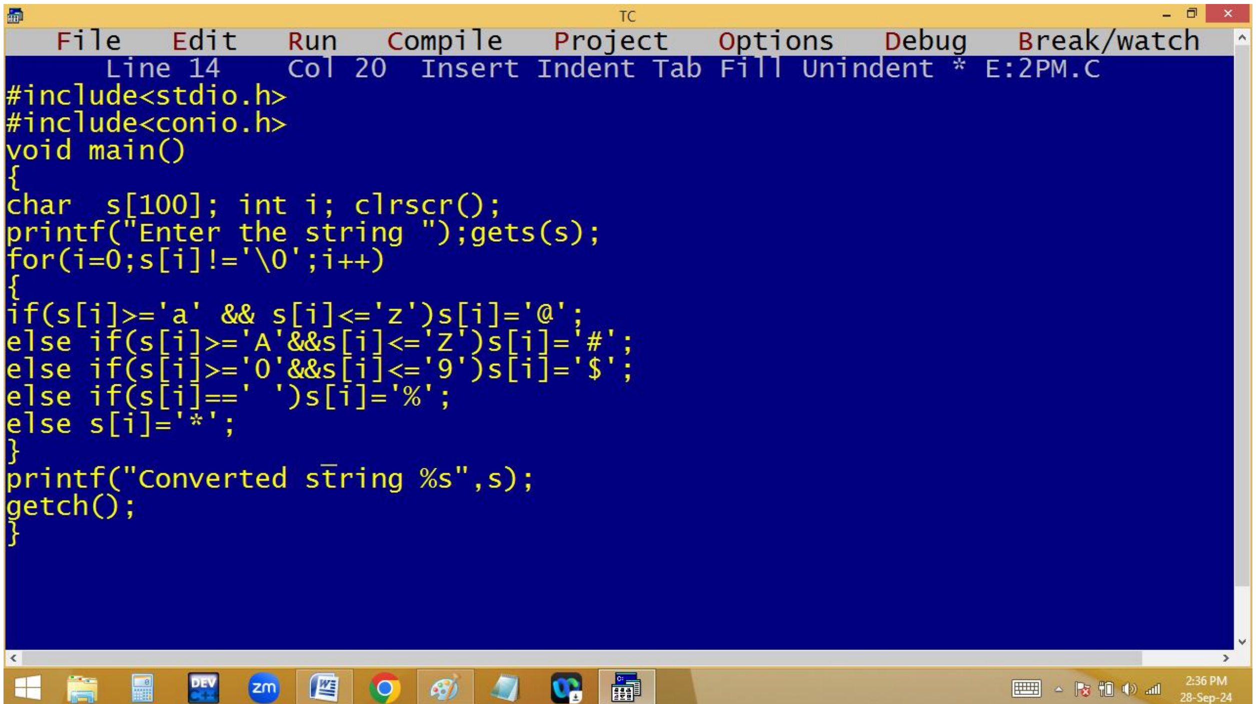
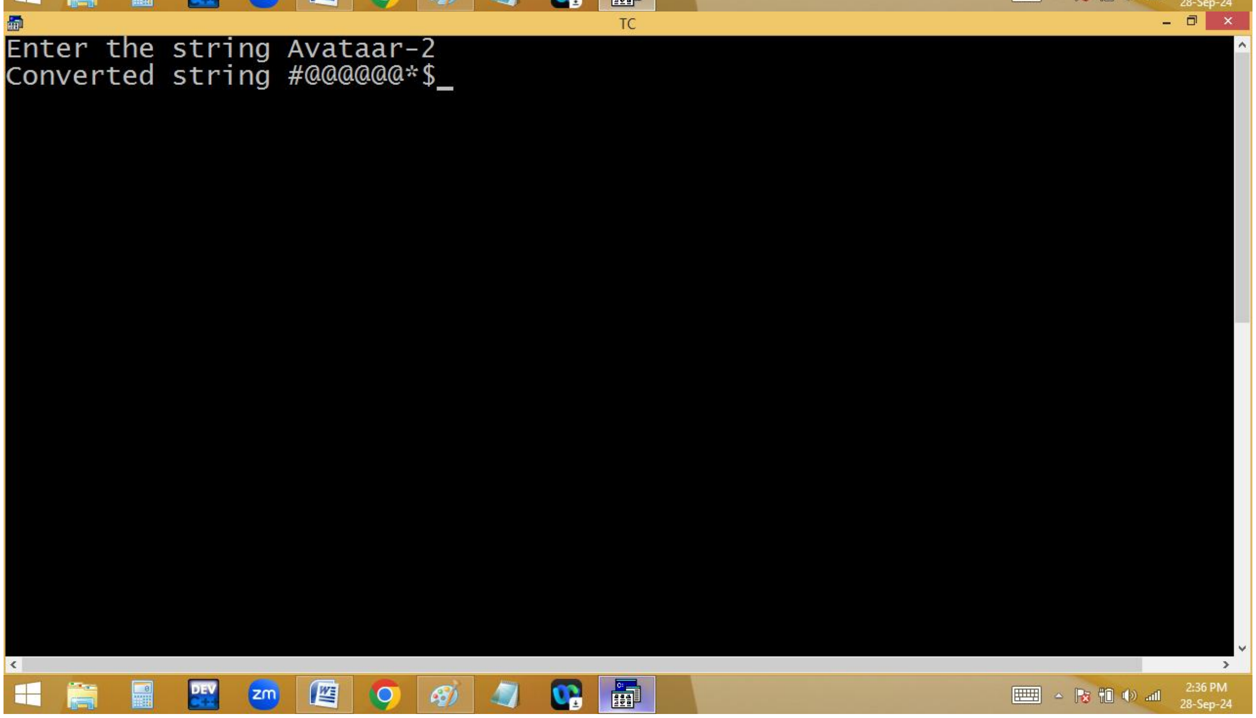
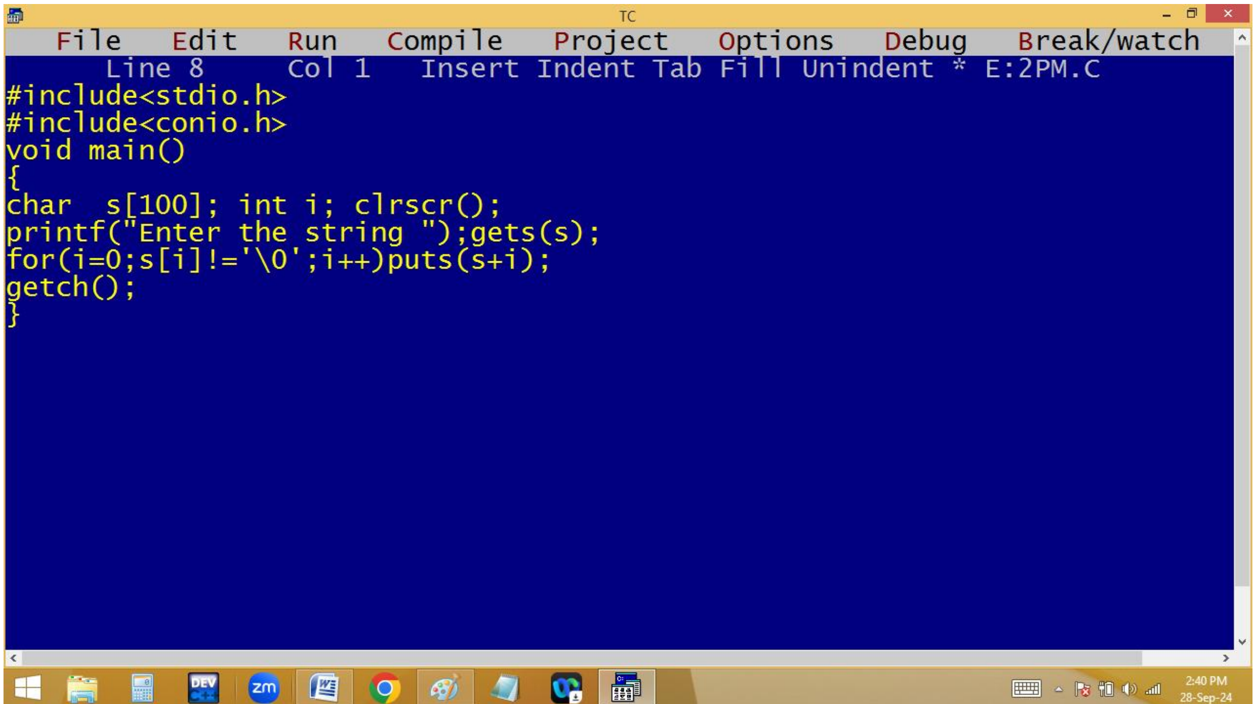




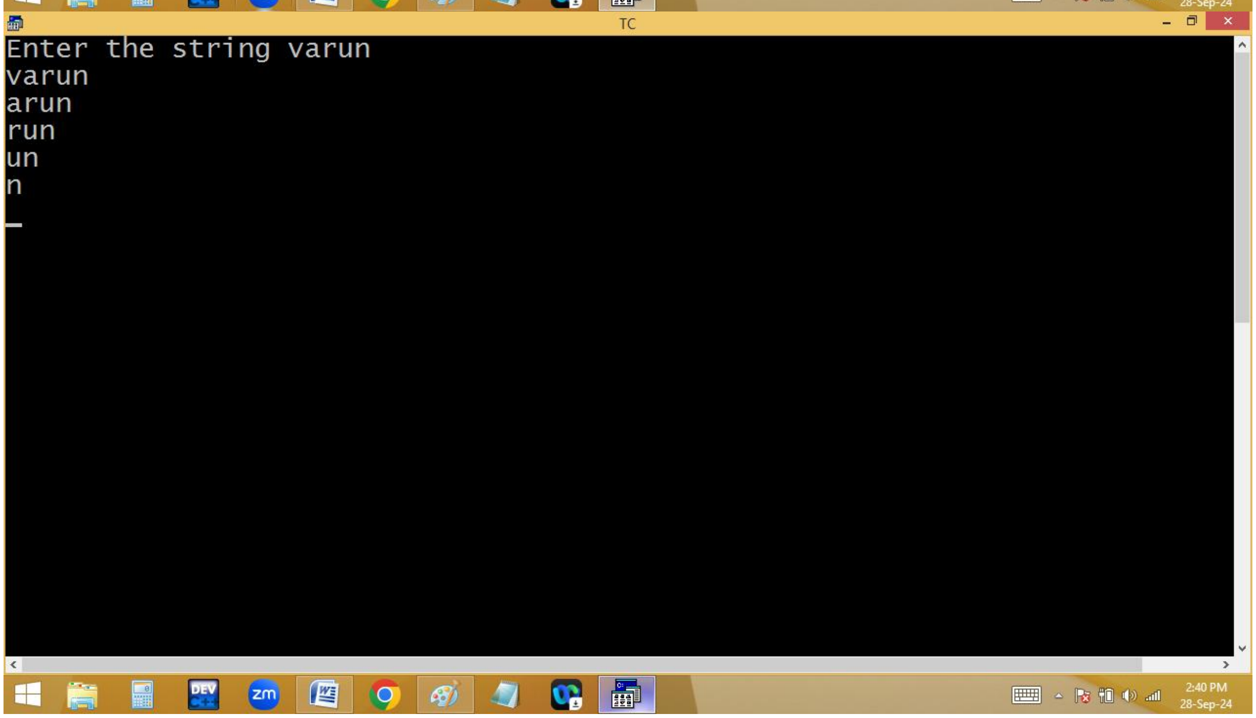
Using ascii values:

```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 13     Col 17  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[100]; int i; clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++)
{
if(s[i]>=97 && s[i]<=122)s[i]='@';
else if(s[i]>=65&&s[i]<=90)s[i]='#';
else if(s[i]>=48&&s[i]<=57)s[i]='$';
else if(s[i]==32)s[i]='%';
else s[i]='*';
}
printf("Converted string %s",s);
getch();
}
```

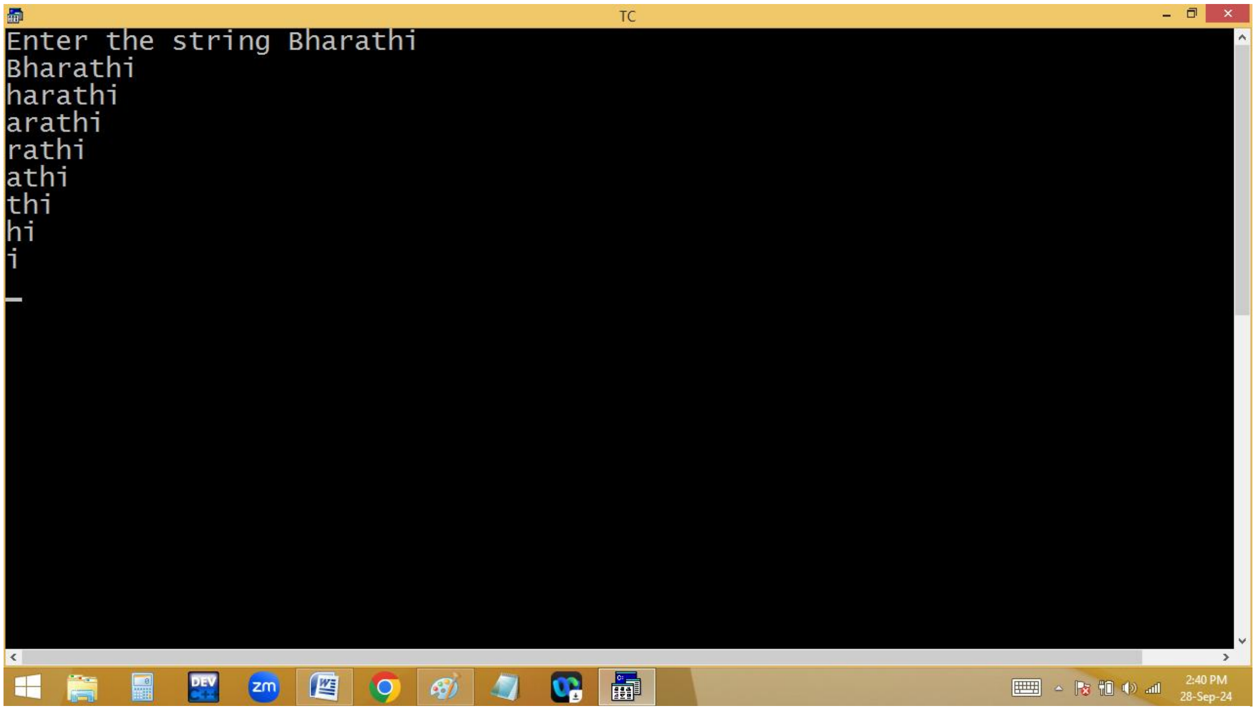
```
Enter the string Bablu aur Bunty - 2
Converted string #@@@@%@@@%#@@@@%*%$
```

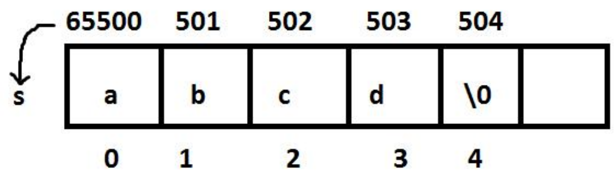


```
TC
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
      Line 8     Col 1    Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[100]; int i; clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++)puts(s+i);
getch();
}
```



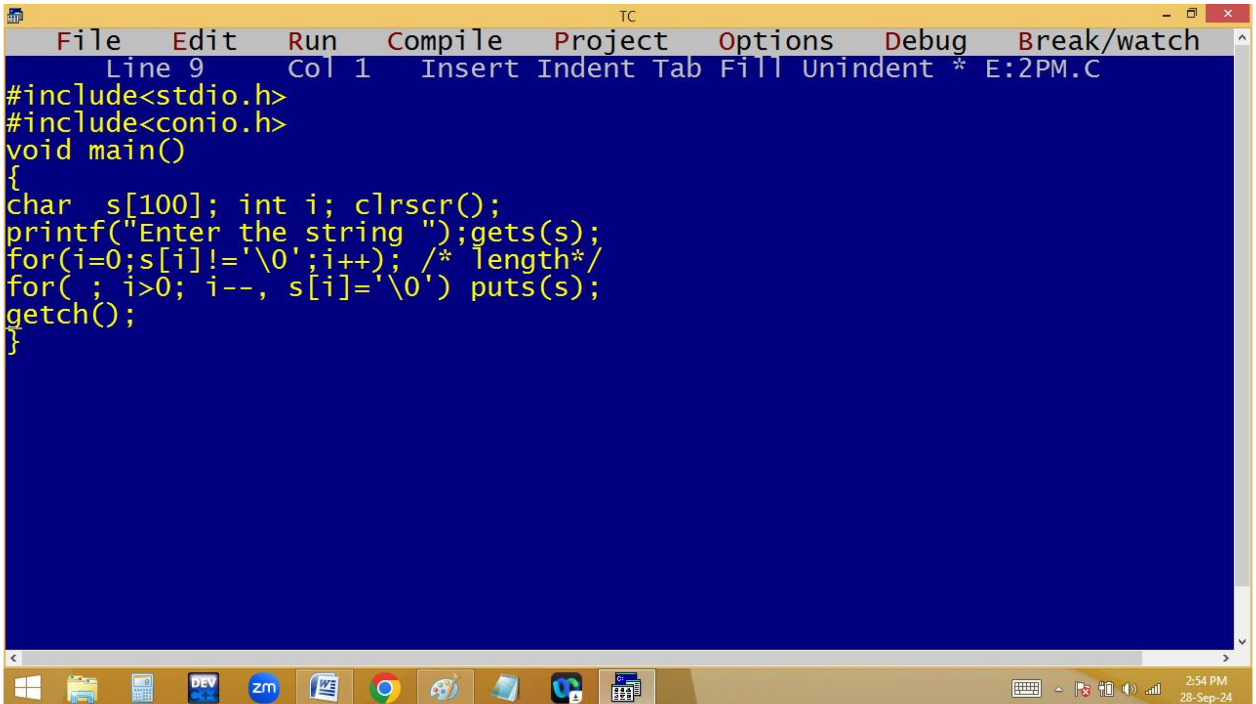
```
TC
Enter the string varun
varun
arun
run
un
n
```

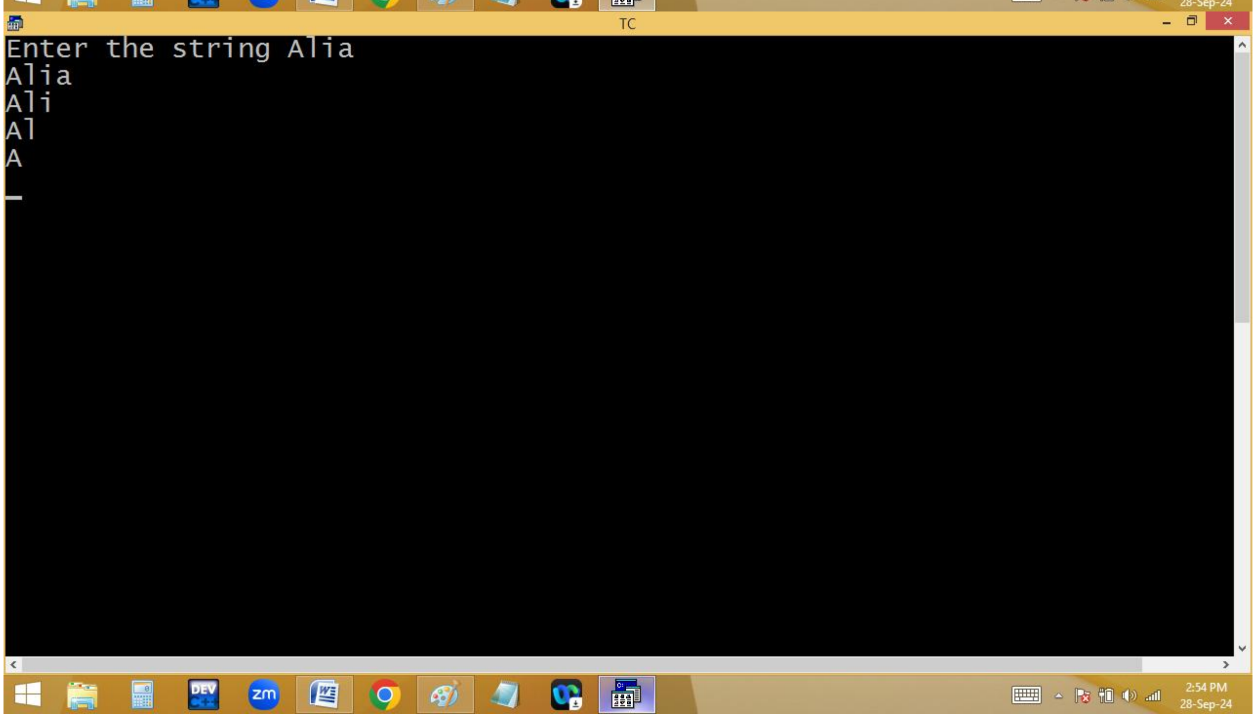




for(i=0; s[i]!='\0'; i++) puts(s+i);

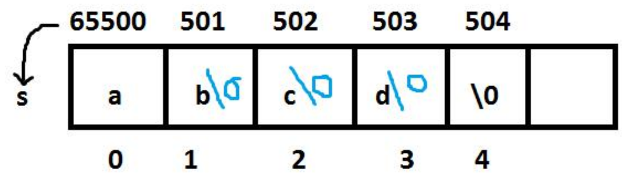
65500+0*1=65500 to \0 ==> abcd
65500+1*1=65501 to \0 ==> bcd
65500+2*1=65502 to \0 ==> cd
65500+3*1=65503 to \0 ==> d
65500+4*1=65504 = '\0' !='\0' ==> false


TC
File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int i; clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++); /* length*/
for(; i>0; i--, s[i]='\0') puts(s);
getch();
}
2:54 PM
28-Sep-24

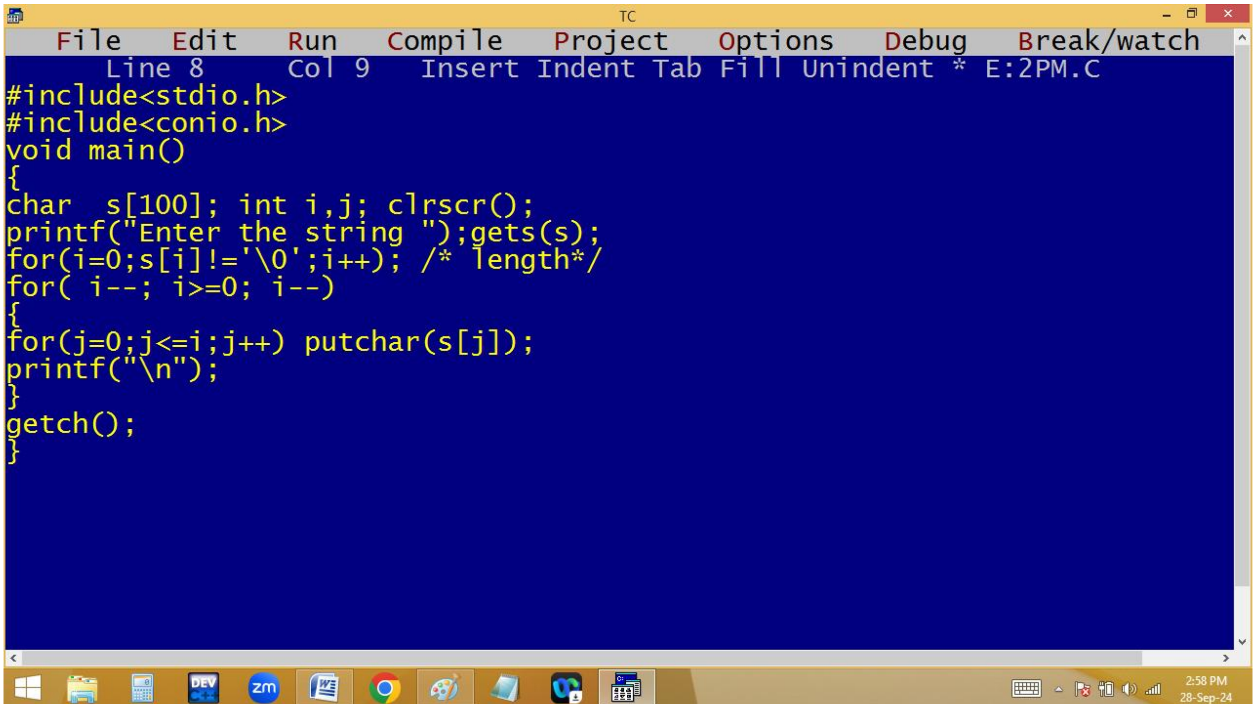

TC
Enter the string Alia
Alia
Ali
Al
A
2:54 PM
28-Sep-24

for(i=0; s[i]!='\0';i++); length=4

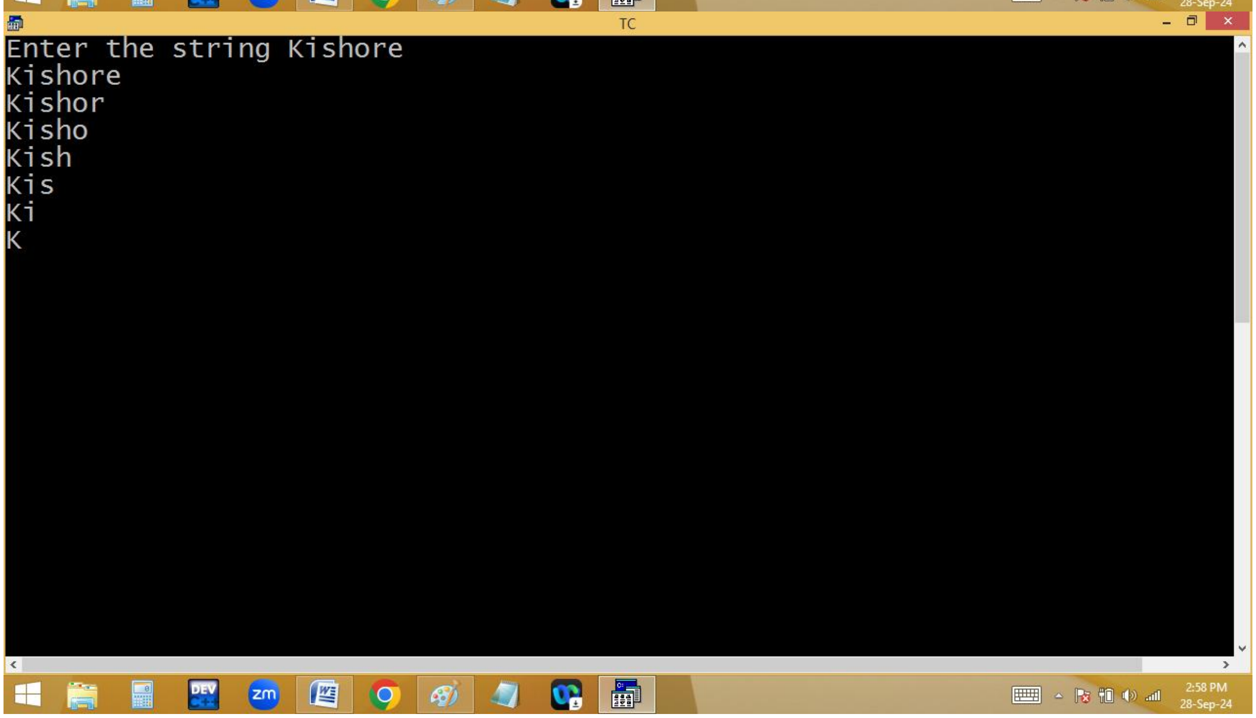
for(; i>0; i--, s[i]='\0') puts(s);



i		
0 – 4	→	abcd
0 – 3	→	abc
0 – 2	→	ab
0 – 1	→	a
0		

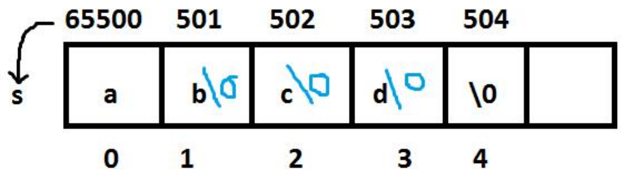



```c
#include<stdio.h>
#include<conio.h>
void main()
{
char  s[100]; int i,j; clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++); /* length*/
for( i--; i>=0; i--)
{
for(j=0;j<=i;j++) putchar(s[j]);
printf("\n");
}
getch();
}
```




```
Enter the string Kishore
Kishore
Kishor
Kisho
Kish
Kis
Ki
K
```

```
for( i=0; s[i]!='\0';i++); length=4
      i--
for( ; i>=0; i-- )
{
for( j=0; j<=i; j++) p(s[ j]);
p("\n");
}
```



j		i
0	—	4
0	—	3
0	—	2
0	—	1

abcd
abc
ab
a

Finding palindrome or not using single string:

liril, madam, malayalam,…

TC

File Edit Run Compile Project Options Debug Break/watch

Line 12 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char  s[100]; int i,j; clrscr();
printf("Enter the string ");gets(s);
for(i=0;s[i]!='\0';i++); /* length*/
for( j=0; j<i/2; j++ )
{
if(tolower(s[j])!=tolower(s[i-j-1]))
{puts("Not a Palindrome");getch();return;}
}
puts("Palindrome");
getch();
}
```

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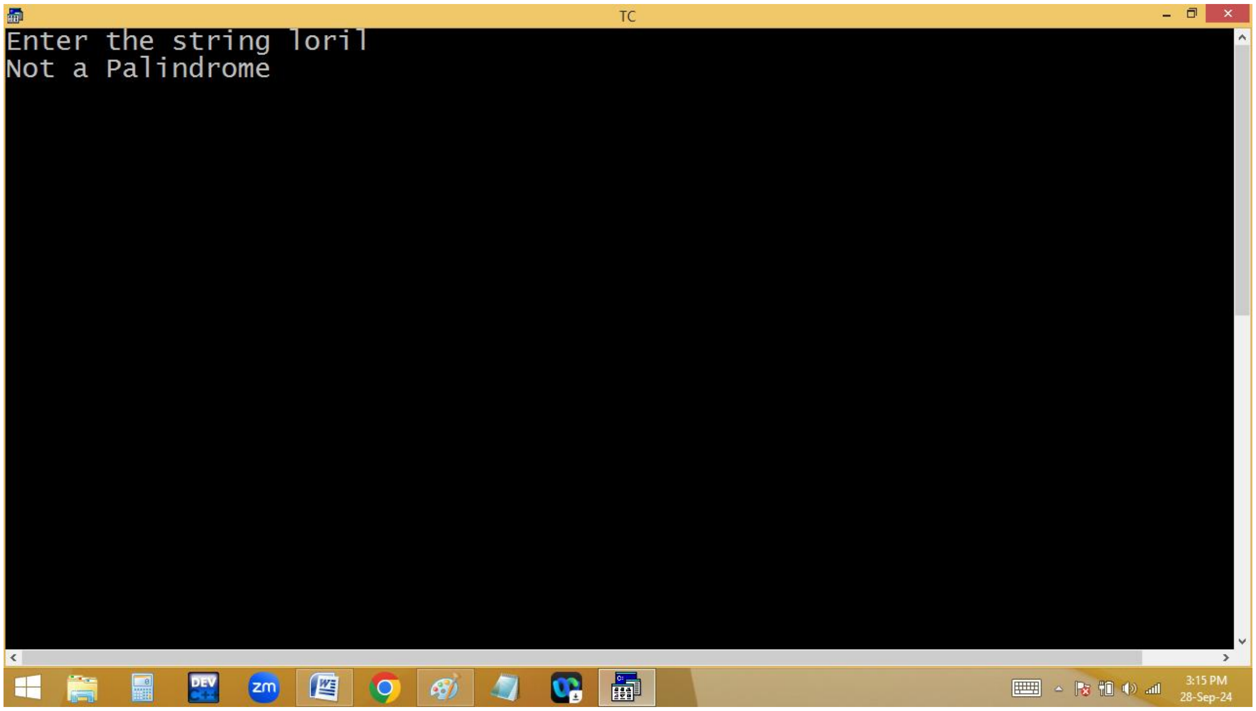
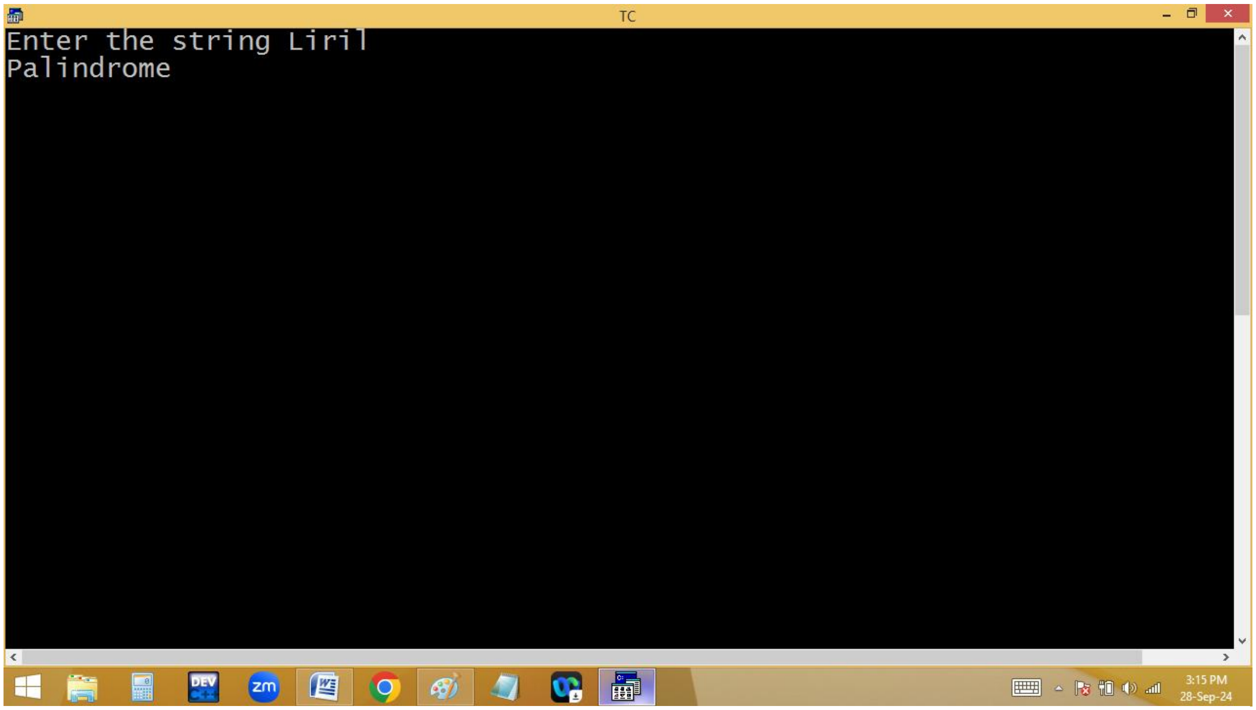
28-Sep-24

TC

```
Enter the string Mom
Palindrome
```

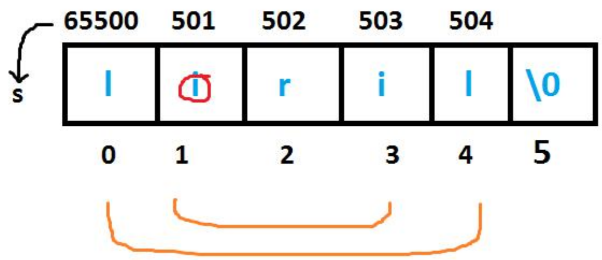
3:15 PM
28-Sep-24


TC
Enter the string loril
Not a Palindrome
3:15 PM
28-Sep-24




```
for( i=0; s[i]!='\0';i++ ) ;  len=i=5

for( j=0; j<i/2;j++ )
{
if(s[ j]!= s[i-j-1] )p("not");return;
}
p(palin);
```



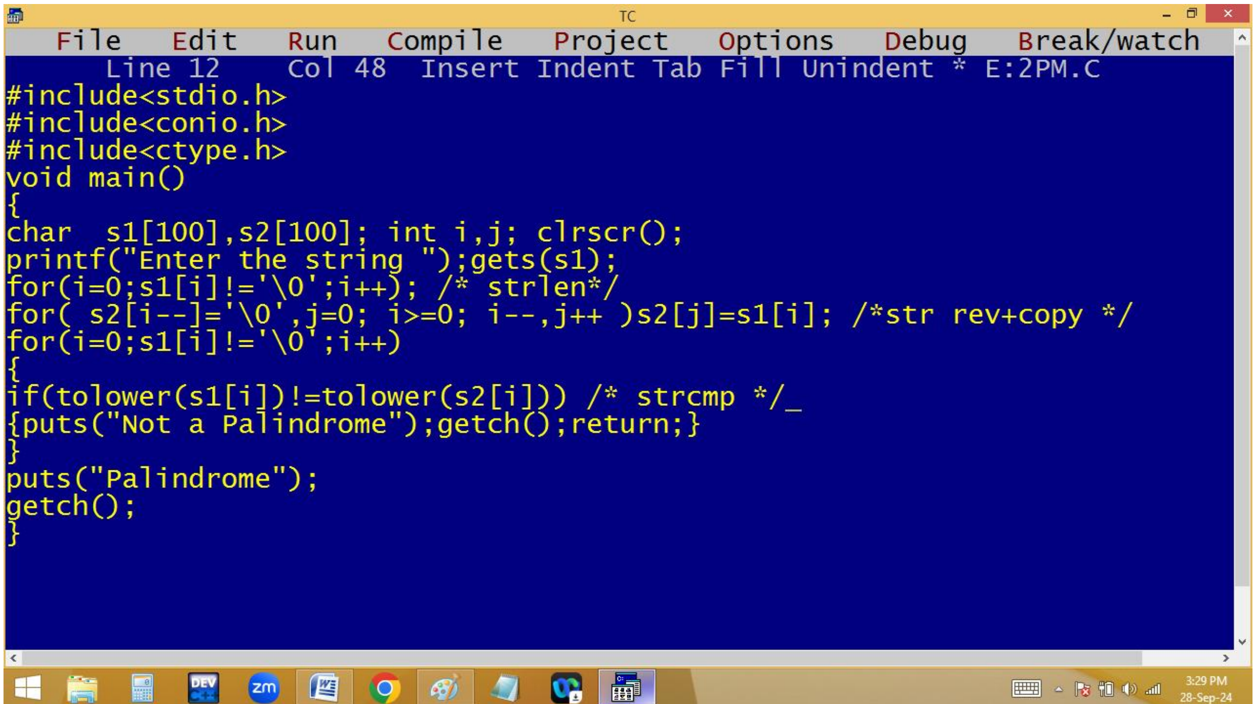
j
0

i
5-0-1=4

1
5-1-1=3

2

Using two strings:



TC

File Edit Run Compile Project Options Debug Break/watch

Line 12 Col 48 Insert Indent Tab Fill Unindent * E:2PM.C

```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char  s1[100],s2[100]; int i,j; clrscr();
printf("Enter the string ");gets(s1);
for(i=0;s1[i]!='\0';i++); /* strlen*/
for( s2[i--]='\0',j=0; i>=0; i--,j++ )s2[j]=s1[i]; /*str rev+copy */
for(i=0;s1[i]!='\0';i++)
{
if(tolower(s1[i])!=tolower(s2[i])) /* strcmp */_
{puts("Not a Palindrome");getch();return;}
}
puts("Palindrome");
getch();
}
```

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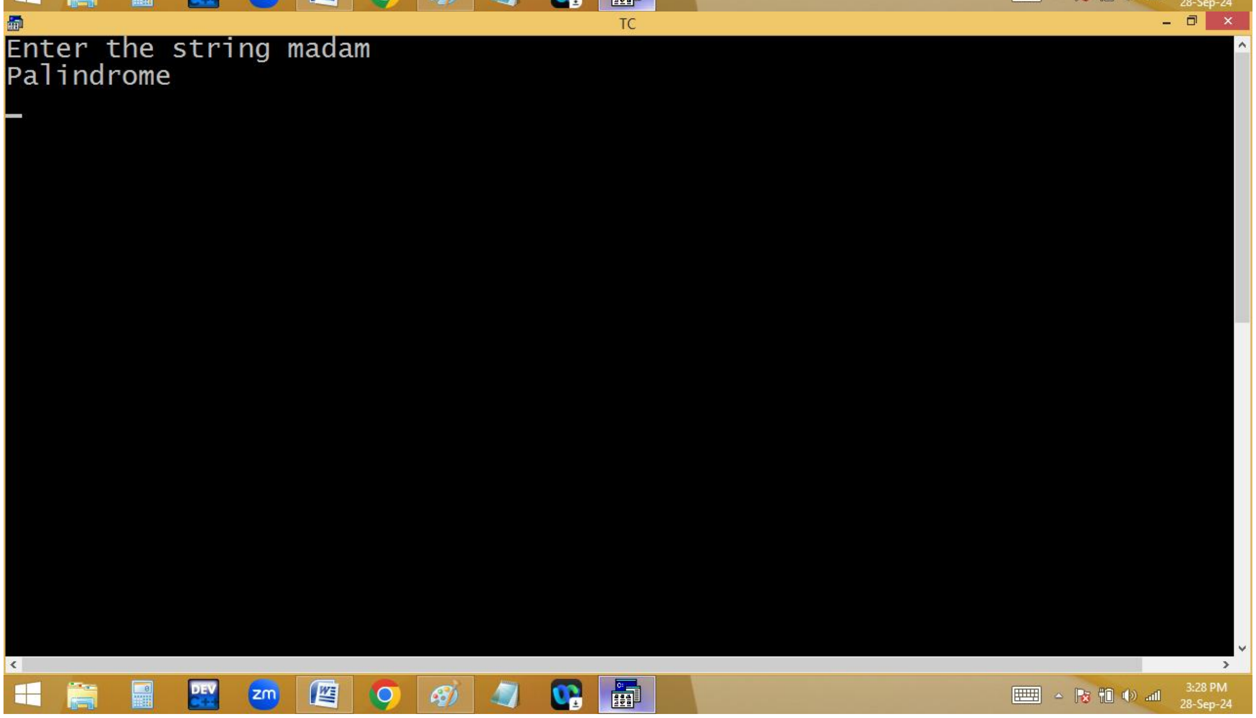
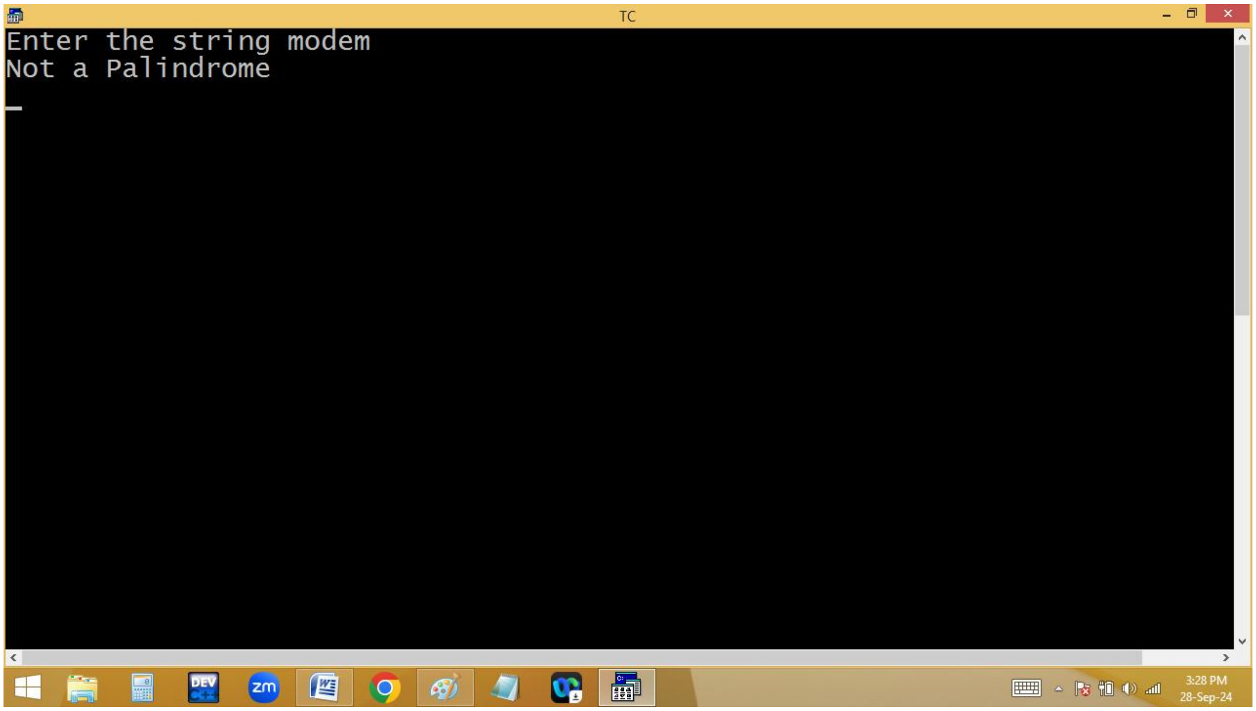
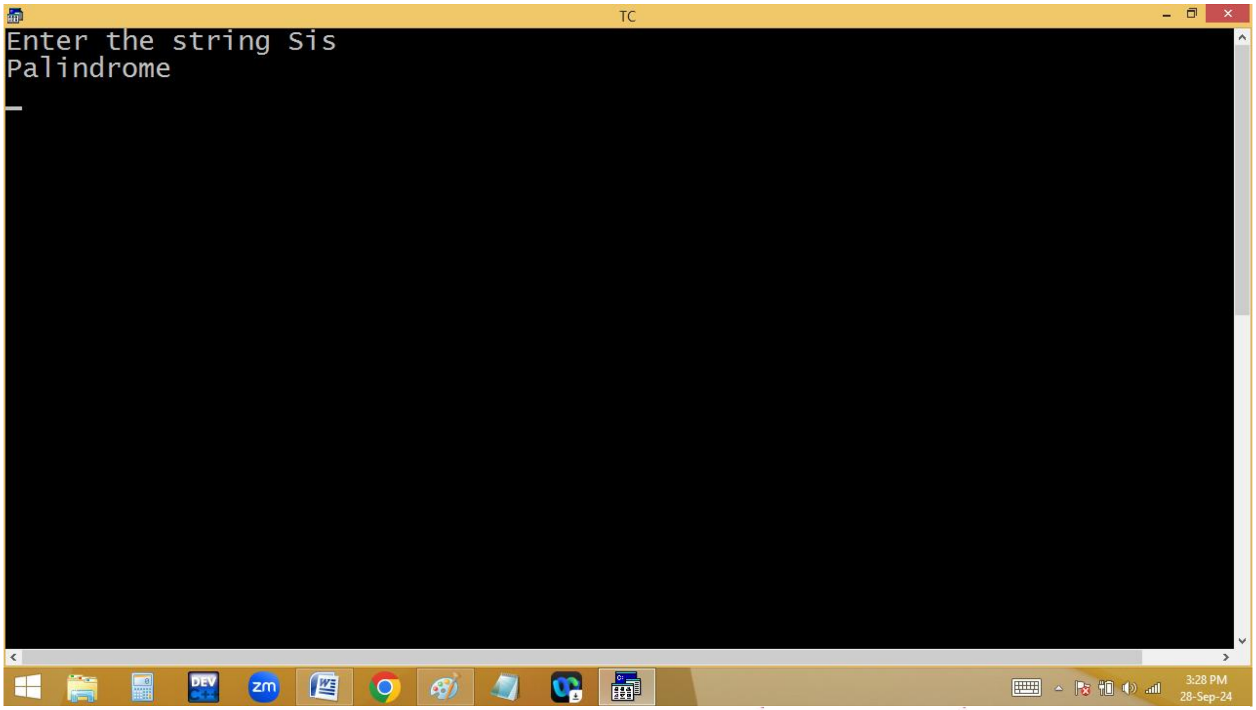


TC

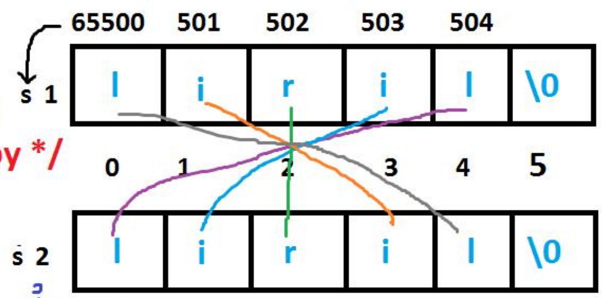
```
Enter the string madam
Palindrome
_
```

3:28 PM 28-Sep-24


TC
Enter the string modem
Not a Palindrome
3:28 PM
28-Sep-24



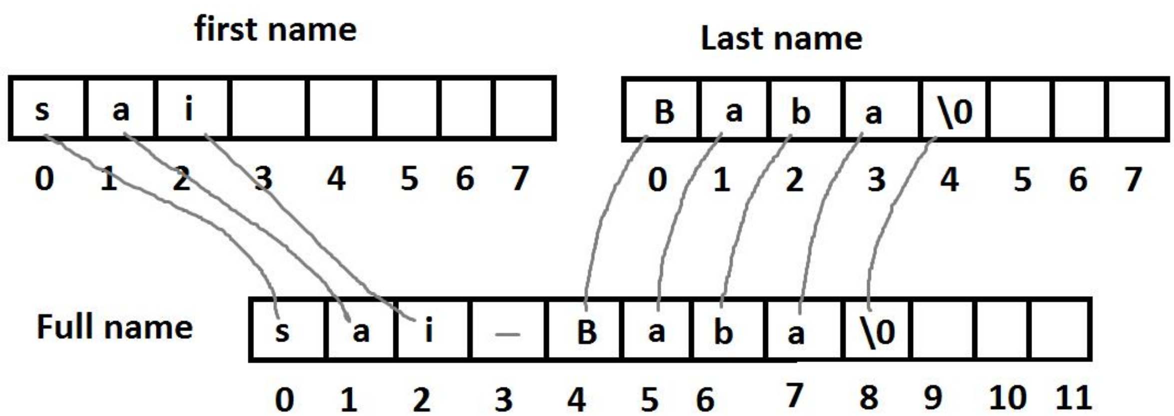

```
for( i=0;s1[i]!='\0';i++ ) ;  len=i=5
                                strlen
for( s2[i--]='\0', j=0;i>=0;i--,j++)s2[ j]=s1[i];
                        /* str rev + copy */

for(i=0;s1[i]!='\0';i++)
{
if(s1[i]!=s2[i])p(not);return;
}   /*strcmp*/
p(palin);
```



i--		j++
4	→	0
3	→	1
2	→	2
1	→	3
0	→	4
-1		

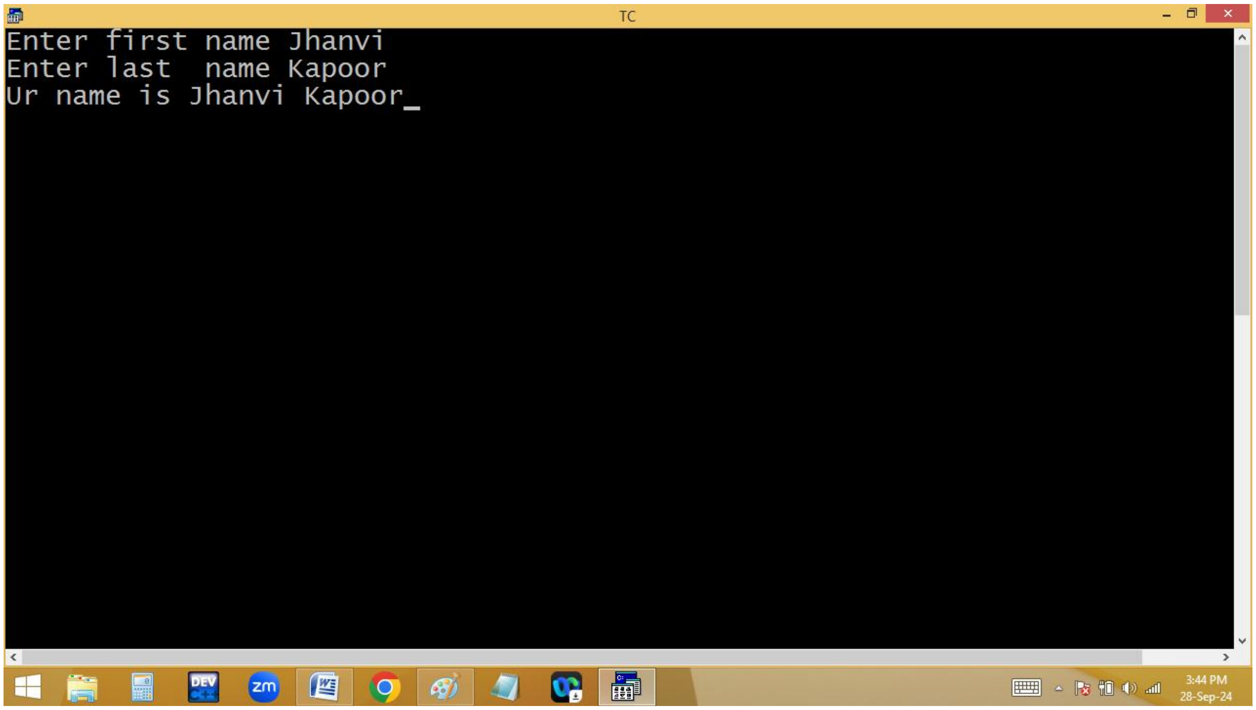
String concatenation with space:

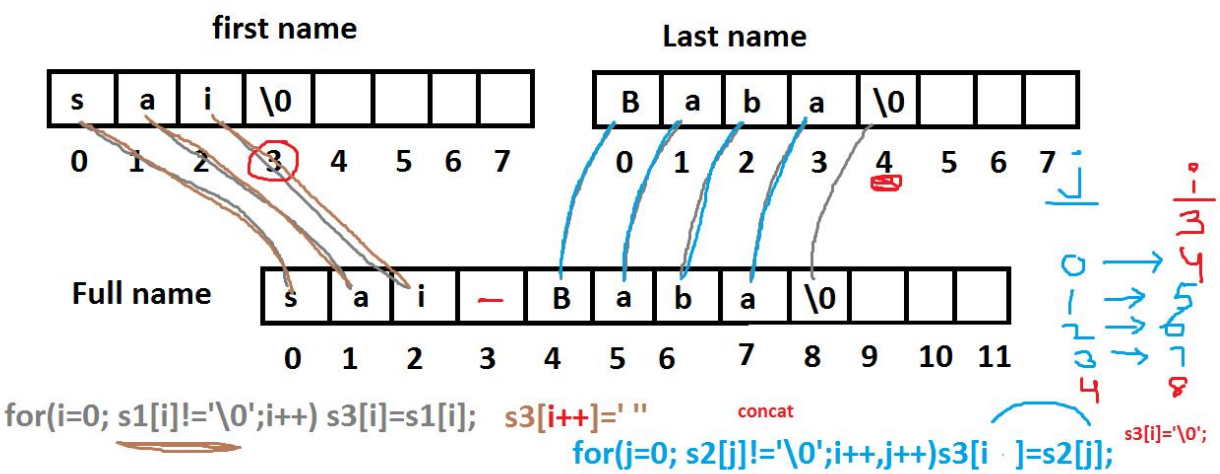
Enter first name: Kishore
Enter last name: Naidu
Ur name is Kishore Naidu



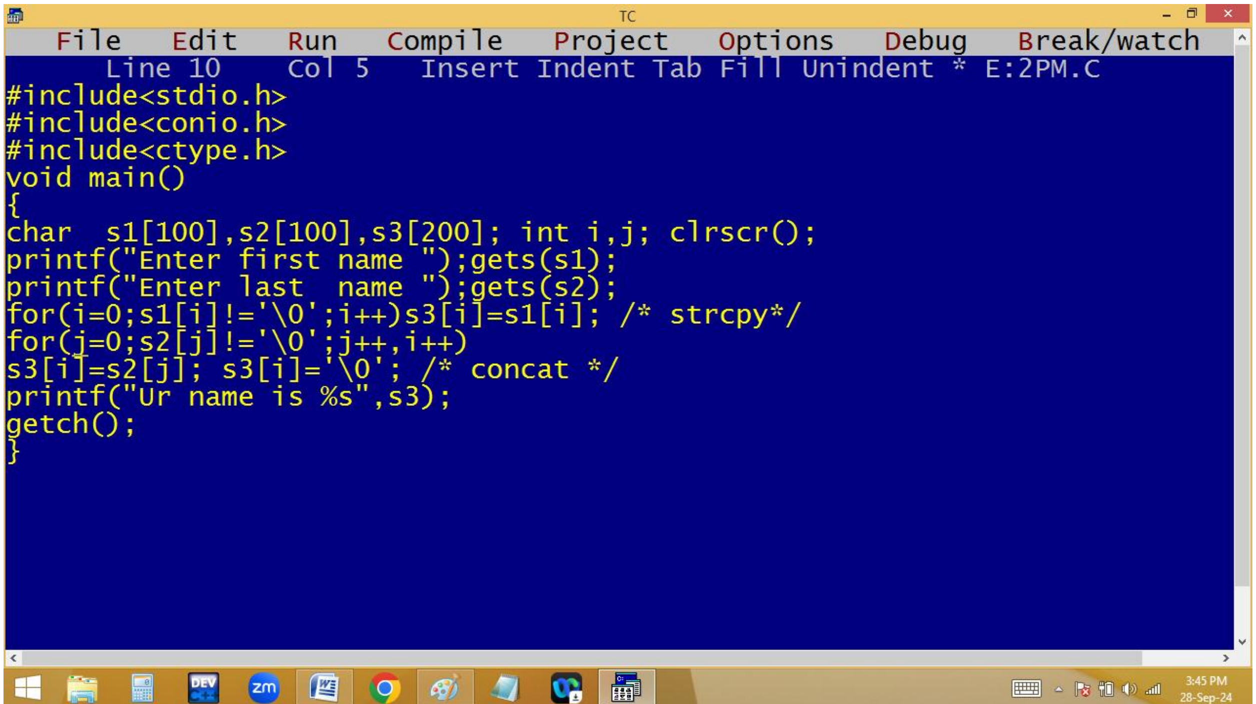
```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 15    Col 1   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char  s1[100],s2[100],s3[200]; int i,j; clrscr();
printf("Enter first name ");gets(s1);
printf("Enter last  name ");gets(s2);
for(i=0;s1[i]!='\0';i++)s3[i]=s1[i]; /* strcpy*/
for(s3[i++]=' ', j=0;s2[j]!='\0';j++,i++)
s3[i]=s2[j]; s3[i]='\0'; /* concat */
printf("Ur name is %s",s3);
getch();
}
```

```
Enter first name Sai
Enter last  name Baba
Ur name is Sai Baba_
```

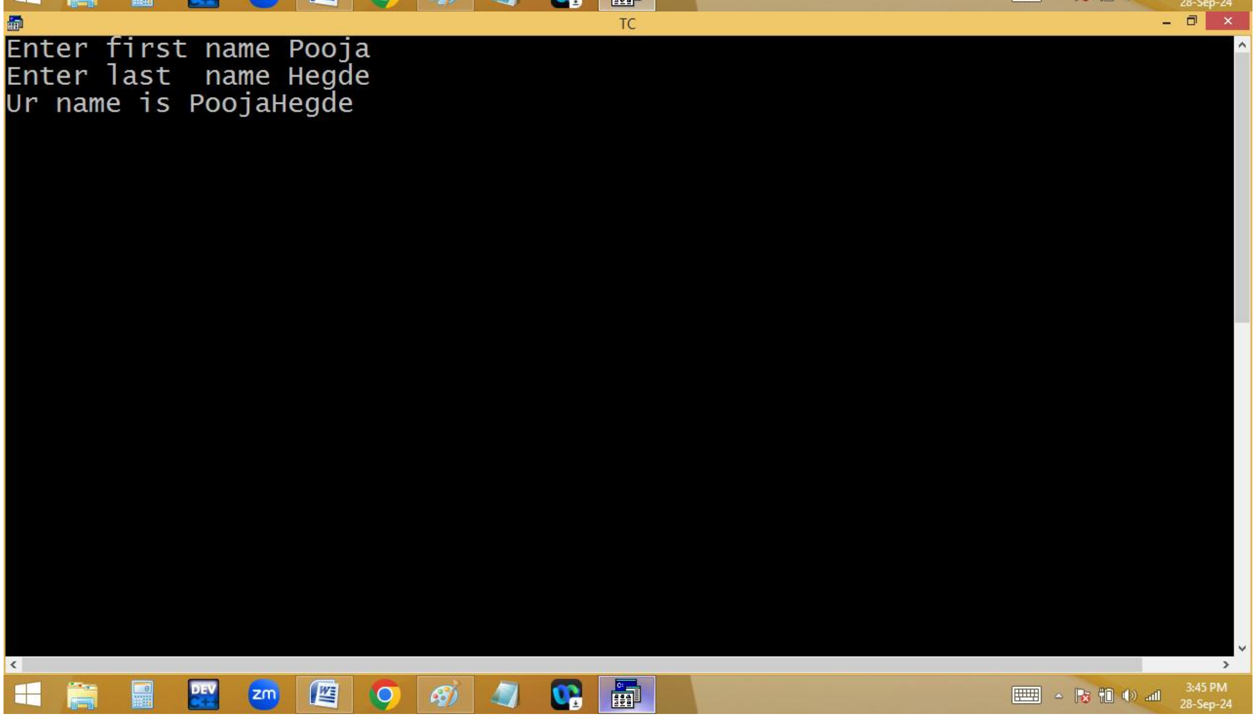




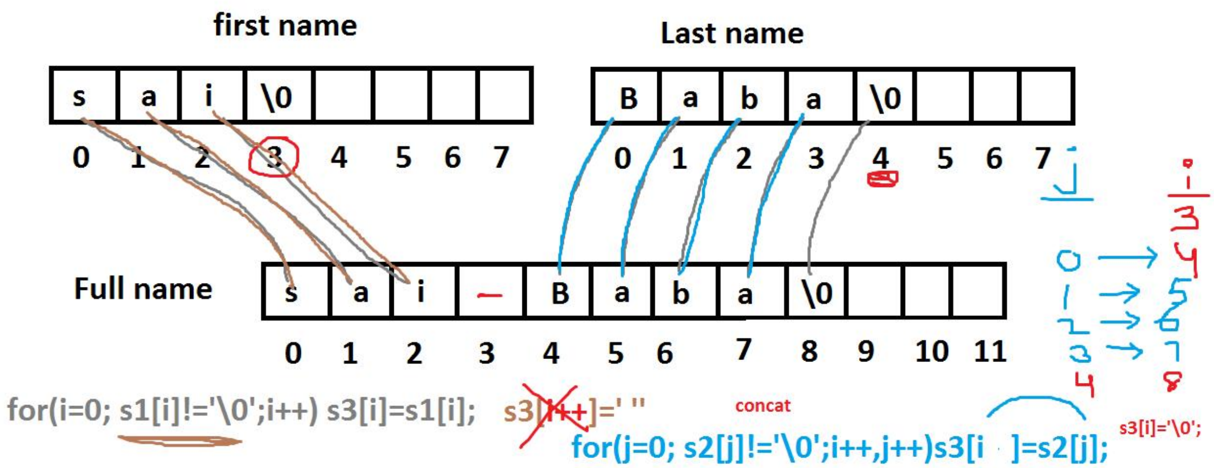
Without space:



```
File  Edit  Run  Compile  Project  Options  Debug  Break/watch
    Line 10    Col 5   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char  s1[100],s2[100],s3[200]; int i,j; clrscr();
printf("Enter first name ");gets(s1);
printf("Enter last  name ");gets(s2);
for(i=0;s1[i]!='\0';i++)s3[i]=s1[i]; /* strcpy*/
for(j=0;s2[j]!='\0';j++,i++)
s3[i]=s2[j]; s3[i]='\0'; /* concat */
printf("Ur name is %s",s3);
getch();
}
```



```
Enter first name Pooja
Enter last  name Hegde
Ur name is PoojaHegde
```



Home work:

Eg1: finding no of words in given sentence.

Jaanu i miss you – 4 words

Eg:2 I am prOUd to be indIAN ➔ I Am Proud To Be Indian.

Eg3: Kabhi Kushi Kabhi Ghum ➔ KKKG

Ranam Roudram Rudhiram ➔ R R R